ISMS

Project Software Requirements Analysis

Team members

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TASK 1

Business Analysis & Problem Understanding

1.1 Current Situation (As-Is Analysis)

Future Vision International School (FVIS) is a private K–12 institution experiencing rapid growth in student enrollment, staff recruitment, and academic program expansion. Despite this growth, the school continues to operate its core administrative and academic functions through a combination of manual paperwork, disconnected Microsoft Excel spreadsheets, and isolated software tools that do not communicate with one another.

The following describes the current operational state across each major function:

Student Records Management

Student personal, academic, and guardian information is maintained in physical folders and multiple unlinked Excel workbooks. There is no single source of truth. Retrieving a student record requires manual searching across multiple files or cabinets, leading to delays and frequent data inconsistencies between departments.

Attendance Tracking

Classroom attendance is recorded manually on paper sheets by teachers at the start of each period. These sheets are collected weekly by the administration and manually entered into spreadsheets. There is no real-time visibility for parents or administrators, and late or absent entries are common.

Grade & Exam Management

Teachers record student grades for assignments, quizzes, and examinations in individual Excel sheets. Grade aggregation and GPA calculation are performed manually, making the process error-prone and time-consuming. Report cards are typed and printed individually by administrative staff at the end of each term.

Fee Collection & Finance

School fees are collected in person at a physical finance counter. Receipts are handwritten or printed from a basic template. Monthly reconciliation is performed manually by the finance team, which frequently results in discrepancies. There is no online payment option and no automated tracking of outstanding balances.

Parent Communication

Communication between the school and parents is conducted through printed circulars distributed in students' bags, occasional phone calls, and informal messaging applications. There is no structured, trackable communication channel. Important notifications such as absences or grade reports often reach parents with significant delays.

Scheduling

Class timetables and exam schedules are created manually by administrators using paper charts and Excel. Scheduling conflicts (double-booked rooms or teachers) are identified only after the schedule is distributed, requiring time-consuming revisions.

Reporting & Decision Support

School leadership has no access to real-time or consolidated data. Any management report requires a staff member to manually gather data from multiple sources, compile it, and present it — a process that takes days and often results in outdated insights by the time decisions are made.

1.2 Business Problems & Inefficiencies

The following table documents the identified business problems, their root causes, measurable impacts, and the stakeholder groups most affected. These problems collectively justify the investment in a centralized Integrated School Management System.

#	Problem	Root Cause	Impact	Affected Stakeholders
1	**No centralized student data repository**	Multiple disconnected Excel files and physical folders	Data duplication, retrieval delays, inconsistent records across departments	Administrators, Teachers, Finance
2	**Manual attendance recording and reporting**	Paper-based system with no digital integration	Errors in records, no real-time visibility, delayed parent notifications	Teachers, Parents, Administrators
3	**Error-prone grade management**	Individual teacher spreadsheets with manual GPA calculation	Inaccurate grades, delayed report cards, no academic trend visibility	Students, Teachers, Parents, Principal
4	**Inefficient manual fee collection**	No payment system; cash-only counter transactions	Reconciliation errors, fraud risk, no payment history tracking for parents	Finance, Parents, Administrators
5	**Poor and unstructured parent communication**	Reliance on physical circulars and informal apps	Delayed critical notifications, low parent engagement, no audit trail	Parents, Teachers, Administrators
6	**No strategic reporting or dashboards**	Data siloed across departments with no aggregation layer	Leadership cannot make data-driven decisions; reports take days to compile	Principal, School Board
7	**Manual and conflict-prone scheduling**	No scheduling engine or conflict detection logic	Timetable conflicts, resource wastage, repeated revisions	Administrators, Teachers, Students
8	**No digital document storage**	Physical filing system	Risk of data loss, slow document retrieval, no version control	Administrators, IT

1.3 Future State — To-Be Business Process Goals

The following goals describe the desired state of school operations following the successful implementation of the ISMS. Each goal directly addresses one or more of the identified business problems and serves as a guiding principle for requirements elicitation in subsequent tasks.

ID	Goal	Description
G-01	**Unified Data Platform**	Establish a single, centralized digital repository for all student, staff, academic, and financial data — eliminating data silos and ensuring a single source of truth across all departments.
G-02	**Automated Attendance Management**	Enable teachers to record attendance digitally in real time, with automated parent notifications triggered immediately upon a student's absence.
G-03	**Streamlined Academic Management**	Provide a fully integrated academic module for grade entry, automatic GPA calculation, report card generation, and exam scheduling with built-in conflict detection.
G-04	**Digital Fee Management**	Replace cash-based fee collection with an online payment system featuring automated invoice generation, digital receipts, payment reminders, and real-time financial dashboards.
G-05	**Structured Parent Engagement**	Deliver a dedicated parent portal providing real-time visibility into student performance, attendance, schedules, and fee status, alongside a structured messaging channel with teachers and staff.
G-06	**Data-Driven Leadership**	Equip the Principal and School Board with live dashboards and customizable reports presenting enrollment trends, academic performance metrics, financial summaries, and operational KPIs.
G-07	**Role-Based Access & Security**	Enforce strict role-based access control ensuring every user interacts only with data and functions relevant to their role, protecting student data privacy and institutional confidentiality.
G-08	**Scalable & Reliable Infrastructure**	Design the system to handle current needs while supporting future growth up to 5,000 students, with high availability, automated backups, and disaster recovery capabilities.

1.4 Business Objectives & Success Criteria

Business objectives translate the To-Be goals into measurable, time-bound targets. Each objective is paired with a concrete success criterion and a target value to enable post-implementation evaluation.

ID	Business Objective	Success Criterion	Target	Timeline
OBJ-01	Eliminate all paper-based student records	100% of student records migrated to the ISMS digital database	**100%**	Go-Live
OBJ-02	Reduce attendance recording errors	Attendance data accuracy rate measured via audit sampling	**≥ 99%**	3 months post go-live
OBJ-03	Automate fee processing	Reduction in manual finance staff hours spent on fee collection and reconciliation	**≥ 60%**	6 months post go-live
OBJ-04	Accelerate grade reporting	Time from exam completion to published report card	**≤ 48 hours**	From Term 1
OBJ-05	Increase parent engagement	Percentage of registered parents actively using the parent portal monthly	**≥ 80%**	6 months post go-live
OBJ-06	Enable real-time decision-making	Frequency of dashboard usage by Principal and Board	**Weekly usage**	From Go-Live
OBJ-07	Achieve high system availability	System uptime during operational hours (7 AM – 9 PM, school days)	**≥ 99.5%**	Ongoing
OBJ-08	Ensure scheduling efficiency	Number of timetable conflicts detected after schedule publication	**0 conflicts**	From Term 1

Task 1 Summary

Task 1 has established the full business context for the ISMS project: the current broken state of FVIS operations, the 8 root-cause problems driving this initiative, the 8 To-Be goals defining the target state, and 8 measurable objectives with clear success criteria. Every requirement, diagram, and model produced by the team in Tasks 2 through 8 must trace back to this context.

TASK 2
Stakeholders Identification & Analysis

2.1 Stakeholder Identification

Stakeholder Group	*Specific Stakeholder*	*Role in the Project*
Internal – Management	School Board	Strategic decision-makers, project sponsors
	Principal / Director	Daily operational leader, primary decision-maker
	Head of Academics	Oversees academic modules (grades, exams, classes)
	Head of Finance	Manages fee collection, invoices, financial reporting
	IT Manager	Responsible for infrastructure, security, deployment
Internal – Staff	Teachers	Use system for attendance, grade entry, parent communication
	Administrative Staff	Registration, scheduling, reporting, clerical tasks
	Finance Officers	Process fees, reconcile payments, generate receipts
Internal – Students	Students	Access grades, attendance, timetable, assignments
External	Parents / Guardians	View child's progress, receive notifications, pay fees online
	Payment Gateway Provider	Third-party service for online fee transactions
	Ministry of Education (future)	May require data integration (out of scope for v1.0)
Project Team	Business Analyst (Me)	Requirements elicitation, documentation, stakeholder management
	System Architect	Designs system structure and technology stack
	Developer	Implements the system
	Tester	Validates requirements and system quality

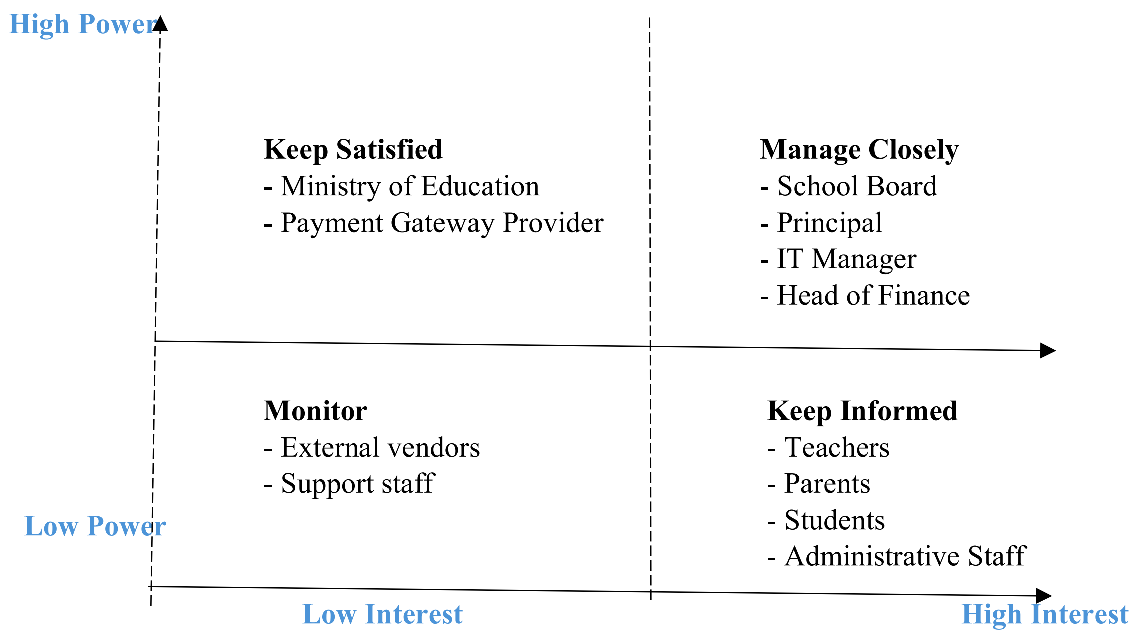
2.2 Stakeholder Roles, Concerns, and Expectations

Stakeholder	*Roles / Responsibilities*	*Key Concerns*	Expectations from ISMS
School Board	Approve budget, set strategic direction	ROI, reputation, compliance, long-term scalability	Dashboards, strategic KPIs, enrollment trends, financial summaries
Principal	Day-to-day operations, leadership	Real-time visibility, ease of use, minimal disruption	Live dashboards, conflict-free scheduling, automated reporting
Head of Academics	Curriculum, exams, grading	Accuracy of grades, smooth exam scheduling, report card quality	Automated GPA, exam conflict detection, grade analytics
Head of Finance	Fee collection, reconciliation	Fraud prevention, reduced manual errors, payment tracking	Online payments, automated invoicing, financial dashboards
IT Manager	System deployment, maintenance	Security, uptime, backups, integration	Role-based access, audit logs, automated backups, high availability
Teachers	Mark attendance, enter grades, communicate	Simplicity, time-saving, reliability	Quick attendance entry, easy grade upload, parent messaging
Administrative Staff	Student registration, scheduling	Efficiency, reduced paperwork, easy search	Centralized student records, conflict-free scheduling, reports
Finance Officers	Fee processing, reconciliation	Accuracy, audit trail, reduction in cash handling	Digital receipts, payment history, overdue flags
Students	Access their own data	Privacy, clarity, ease of access	View grades, attendance, timetable, assignments
Parents / Guardians	Monitor child's progress, pay fees	Transparency, timely notifications, convenience	Parent portal, absence alerts, fee reminders, report cards

Payment Gateway Provider	Process online transactions	Security, compliance, API reliability	Secure EGP transactions, transaction logs, callback URLs
Business Analyst (Me)	Define requirements, manage scope	Clear communication, traceability, stakeholder alignment	Well-documented, testable, and approved requirements
Developer	Build the system	Clear specifications, feasible scope	Functional & non-functional requirements, UML diagrams
Tester	Verify system quality	Testable requirements, edge cases	Requirement traceability, acceptance criteria

2.3 Stakeholder Matrix (Power vs. Interest)

Power / Interest Level	*Category*	*Stakeholders*	*Engagement Strategy*
High Power / High Interest	Manage Closely	School Board, Principal, Head of Finance, IT Manager	Regular meetings, approvals, progress reviews, scope validation
High Power / Low Interest	Keep Satisfied	Ministry of Education (future), Payment Gateway Provider	Monitor, inform as needed, ensure compliance and contract terms
Low Power / High Interest	Keep Informed	Teachers, Parents, Students, Administrative Staff	Training, newsletters, user feedback sessions, support channels



TASK 3

Functional Requirements

1. Student Registration & Information Management

This module handles all student-related data in a centralized and organized way.

- Create and manage student profiles including personal details (name, date of birth, contact information).
- Link each student to one or more parents/guardians.
- Upload and store required documents (ID, certificates, etc.).
- Assign a unique ID to each student.
- Enable quick search and filtering of student records.
- Maintain a complete academic and enrollment history for every student.

2. Teacher & Staff Management

This module manages all staff-related information and assignments.

- Manage teacher and staff profiles and information.
- Assign teachers to classes and subjects.
- Provide access to staff records and directory.

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3. Class Enrollment & Academic Structure

This module organizes how students, classes, and subjects are structured.

- Define academic years, semesters, grades, and class sections.
- Assign students to specific classes and sections.
- Assign teachers to subjects and classes.
- Manage class capacity and avoid over-enrollment.
- Generate and manage class schedules (timetables).
- Detect and prevent scheduling conflicts (teachers, rooms, or time overlaps).

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4. Attendance Recording

This module digitizes the attendance process for accuracy and real-time tracking.

- Allow teachers to record attendance for each class session.
- Store attendance records automatically in the system.
- Support editing or correcting attendance with proper permissions.
- Keep a full attendance history for each student.
- Provide attendance summaries and reports for classes and individuals.

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5. Exam & Grade Management

This module supports all academic evaluation processes.

- Create exams, quizzes, and assignments.

- Record and update student grades.
- Automatically calculate totals, averages, and GPA.
- Generate report cards for each student.
- Manage exam schedules and avoid timing conflicts.
- Allow students and parents to view grades through their portals.
- Track academic performance over time.

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6. Fee Payment Processing

This module handles all financial transactions related to student fees.

- Define fee structures based on grade or academic year.
- Automatically generate invoices for each student.
- Support different payment methods (online payment, etc.).
- Track paid, unpaid, and partially paid fees.
- Generate digital receipts for every payment.
- Send reminders for overdue payments.
- Maintain a full payment history for transparency.

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7. Parent & Student Portals and Communication

This module provides a unified platform for communication and access to academic and administrative information for students and parents.

- Students and parents can access key information such as grades, attendance, schedules, and fee status through simple and user-friendly portals.
- The system provides real-time notifications for important events, including absences, grade updates, and school announcements.
- A structured communication channel allows messaging between parents and teachers, sharing academic updates, and keeping a record of all interactions.

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8. Reporting & Dashboards

This module supports decision-making through clear data visualization.

- Provide dashboards for school management with key insights.
- Generate reports for:
  - Student performance
  - Attendance statistics
  - Financial status
- Allow exporting reports (PDF / Excel).
- Display real-time data for better and faster decisions.

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9. Role-Based Access

This module ensures secure access control based on user roles.

- Different access levels for admins, teachers, students, and parents.
- Each user sees only what they are allowed to see.

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10. Notifications & Alerts System

This module handles system-wide alerts and notifications.

- Central system for sending alerts (fees, attendance, grades).
- Supports multiple channels (system notifications, SMS if needed).

11. Audit & Activity Tracking

This module ensures accountability and system monitoring.

- Track all important system actions.
- Maintain detailed logs showing who performed each action.
- Support data backup to ensure information safety and recovery.

12. System Interface & Language Support

This module enhances usability and accessibility for users.

- Support both Arabic and English interfaces.
- Allow users to switch between languages.

TASK 4

Nonfunctional Requirements

1. Student Registration & Information Management

- The system shall retrieve student records within ≤ **2 seconds**. *(Performance)*
- The system shall ensure **no duplicate student IDs**. *(Data Integrity)*
- Student data shall be **encrypted during transmission and storage**. *(Security)*
- The system shall maintain ≥ **99.5% uptime** for accessing records. *(Availability)*
- Only authorized users (Admin/Registrar) shall view or edit student data. *(Security / RBAC)*

2. Teacher & Staff Management

- Staff profiles shall load within ≤ **2 seconds**. *(Performance)*
- Access to staff data shall be **restricted based on roles**. *(Security)*
- The system shall handle **100+ staff records without performance issues**. *(Scalability)*

3. Class Enrollment & Academic Structure

- The system shall **detect scheduling conflicts in real time** before saving. *(Performance + Reliability)*
- The system shall **prevent over-enrollment automatically**. *(Data Integrity)*
- Timetable generation shall complete within ≤ **5 seconds**. *(Performance)*
- Relationships between students, teachers, and classes shall remain **consistent and accurate**. *(Reliability)*

4. Attendance Recording

- Attendance marking shall take ≤ **2 seconds per class**. *(Performance)*
- The system shall ensure ≥ **99% attendance accuracy**. *(Reliability)*
- Attendance data shall be saved **instantly with no loss**. *(Reliability)*
- Only authorized teachers can edit attendance. *(Security)*
- Notifications for absences shall be sent within ≤ **5 seconds**. *(Performance)*

5. Exam & Grade Management

- Grade calculations (totals, GPA) shall be **automatic and immediate**. *(Performance)*
- The system shall ensure **100% accuracy in grades**. *(Reliability)*
- Report cards shall be generated within ≤ **5 seconds per student**. *(Performance)*
- Grade data shall be **secure and role-restricted**. *(Security)*
- The system shall handle large volumes of grades efficiently. *(Scalability)*

6. Fee Payment Processing

- All payments shall be processed using **secure encryption (SSL/TLS)**. *(Security)*
- Payment confirmation shall appear within **≤ 3 seconds**. *(Performance)*
- The system shall ensure **zero data loss in financial transactions**. *(Reliability)*
- Payment records shall be **complete and tamper-proof**. *(Data Integrity)*
- The system shall support **multiple users paying at the same time**. *(Scalability)*

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7. Parent & Student Portals and Communication

- Portal pages shall load within **≤ 3 seconds**. *(Performance)*
- The system shall be **fully responsive on mobile, tablet, and desktop**. *(Usability)*
- Notifications shall be delivered within **≤ 5 seconds**. *(Performance)*
- The system shall support **Arabic (RTL) and English (LTR)**. *(Usability)*
- Messages and data shall be **secure and role-restricted**. *(Security)*

8. Reporting & Dashboards

- Dashboards shall load within **≤ 3 seconds**. *(Performance)*
- Reports shall be generated within **≤ 5 seconds**. *(Performance)*
- Data shall be **updated in real time**. *(Reliability)*
- The system shall handle large datasets efficiently. *(Scalability)*
- Only authorized users (Principal/Admin) can access reports. *(Security)*

- Cross-System Non-Functional Requirements:

Security

- Role-Based Access Control (RBAC) across all modules
- Protection against SQL Injection, XSS, CSRF
- Secure password storage (hashing + salting)

Availability

- System uptime **≥ 99.5%**
- Backup and recovery within **≤ 1 hour**

Scalability

- Support up to **5,000 students**
- Handle high concurrent users

Maintainability

- Modular and well-structured code
- Easy updates without affecting other modules

Compatibility

- Works on Chrome, Edge, Firefox
- Works on mobile, tablet, desktop

TASK 5

Use Case Modeling

1. RECORD AND EDIT ATTENDANCE

PRIMARY ACTOR

Teacher

PRECONDITIONS

- The teacher must be authenticated in the system.
- The class and timetable must already exist.
- Students must already be assigned to the class.

MAIN FLOW

1. The teacher logs into the system.
2. The teacher selects the class session.
3. The system displays the student list.
4. The teacher records attendance for each student.
5. The teacher submits the attendance record.
6. The system automatically stores attendance records.
7. The system updates attendance summaries.

ALTERNATE FLOWS

- If the teacher enters incorrect attendance data, the teacher may edit the attendance before final submission.
- If the system connection fails, the system displays an error message and requests the teacher to retry.

POSTCONDITIONS

- Attendance records are saved successfully.
- Attendance summaries are updated.
- Students and parents may receive absence notifications.

2. CREATE EXAMS, QUIZZES, AND ASSIGNMENTS

PRIMARY ACTOR

Teacher

PRECONDITIONS

- The teacher must be logged into the system.
- The teacher must be assigned to the subject and class.

MAIN FLOW

1. The teacher accesses the assessment management section.
2. The teacher selects the type of assessment.
3. The teacher enters assessment details such as title, date, and instructions.
4. The teacher uploads or creates the assessment content.
5. The teacher assigns the assessment to students.
6. The system stores the assessment and notifies students.

ALTERNATE FLOWS

- If required information is missing, the system prompts the teacher to complete all fields.
- If the scheduled exam conflicts with another exam, the system displays a scheduling conflict warning.

POSTCONDITIONS

- The assessment is successfully created and stored.
- Students can access the assessment.

3. RECORD AND UPDATE STUDENT GRADES

PRIMARY ACTOR

Teacher

PRECONDITIONS

- The teacher must be authenticated.
- Assessments must already exist.
- Students must be enrolled in the class.

MAIN FLOW

1. The teacher opens the grading section.
2. The teacher selects a class and assessment.
3. The teacher enters student grades.
4. The teacher saves the grades.
5. The system automatically calculates totals, averages, and GPA.
6. The system updates student academic performance records.
7. Notifications are sent to students and parents.

ALTERNATE FLOWS

- If invalid grades are entered, the system displays an error message.
- The teacher may edit grades before final approval.

POSTCONDITIONS

- Student grades are stored successfully.
- Academic performance reports are updated.

4. VIEW PERSONAL GRADES AND ACADEMIC PERFORMANCE

PRIMARY ACTOR

Student

PRECONDITIONS

- The student must log into the system.
- Grades must already be available.

MAIN FLOW

1. The student logs into the system.
2. The student selects the academic performance section.
3. The system displays grades, averages, and GPA.
4. The student reviews academic performance.

ALTERNATE FLOWS

- If no grades are available, the system displays a notification message.

POSTCONDITIONS

The student successfully views academic records.

5. VIEW ATTENDANCE RECORD

PRIMARY ACTOR

Student

PRECONDITIONS

- The student must be authenticated.
- Attendance records must already exist.

MAIN FLOW

1. The student accesses the attendance section.
2. The system retrieves attendance data.
3. The system displays attendance records and summaries.
4. The student reviews attendance information.

ALTERNATE FLOWS

- If attendance records are unavailable, the system informs the student.

POSTCONDITIONS

- Attendance information is displayed successfully.

6. VIEW CHILD’S GRADES AND ACADEMIC PERFORMANCE

PRIMARY ACTOR

Parent/Guardian

PRECONDITIONS

- The parent must log into the system.
- The parent account must be linked to the student account.

MAIN FLOW

1. The parent accesses the student performance section.
2. The system retrieves the child's grades and GPA.
3. The system displays academic performance information.
4. The parent reviews the records.

ALTERNATE FLOWS

- If no grades are available, the system displays a message.

POSTCONDITIONS

- The parent successfully views the child's academic performance.

7. COMMUNICATE WITH TEACHERS

PRIMARY ACTOR

Parent/Guardian

PRECONDITIONS

- The parent must be authenticated.
- The teacher must exist in the system.

MAIN FLOW

1. The parent opens the messaging channel.
2. The parent selects the teacher.
3. The parent writes and sends a message.
4. The system delivers the message to the teacher.
5. The teacher may reply through the system.

ALTERNATE FLOWS

- If the messaging service is unavailable, the system displays an error message.

POSTCONDITIONS

- Communication between parent and teacher is stored successfully.

8. CREATE AND MANAGE STUDENT PROFILES

PRIMARY ACTOR

Administrator

PRECONDITIONS

- The administrator must be logged into the system.

MAIN FLOW

1. The administrator opens the student management section.
2. The administrator enters student information.
3. The administrator creates a new student profile.
4. The system assigns a unique student ID.
5. The administrator updates or manages profile information when needed.
6. The system stores the student profile.

ALTERNATE FLOWS

- If required information is missing, the system requests completion of missing data.
- If a duplicate profile exists, the system displays a warning.

POSTCONDITIONS

- Student profiles are successfully created and stored.

9. GENERATE AND MANAGE TIMETABLES

PRIMARY ACTOR

Administrator

PRECONDITIONS

- Classes, teachers, and subjects must already exist.

MAIN FLOW

1. The administrator accesses timetable management.
2. The administrator assigns classes, teachers, and subjects.
3. The system checks scheduling conflicts.
4. The administrator finalizes the timetable.
5. The system publishes the timetable.

ALTERNATE FLOWS

- If scheduling conflicts are detected, the system prevents timetable submission.
- The administrator may edit the timetable before publishing.

POSTCONDITIONS

- Timetables are generated successfully.
- Students and teachers can view schedules.

10. GENERATE INVOICES FOR STUDENTS

PRIMARY ACTOR

Administrator

PRECONDITIONS

- Fee structures must already be defined.
- Student accounts must exist.

MAIN FLOW

1. The administrator selects the invoice generation option.
2. The administrator chooses the student or class.
3. The system calculates fees based on the defined fee structure.
4. The system generates invoices automatically.
5. The invoices are stored in the system.
6. Notifications are sent to parents and students.

ALTERNATE FLOWS

- If fee structures are not defined, the system prevents invoice generation.
- If student records are incomplete, the system displays an error.

POSTCONDITIONS

- Student invoices are generated and stored successfully.

11. TRACK PAID, UNPAID, AND PARTIALLY PAID FEES

PRIMARY ACTOR

Administrator

PRECONDITIONS

- Student invoices must already exist.

MAIN FLOW

1. The administrator accesses the payment tracking section.
2. The system retrieves payment information.
3. The administrator reviews paid, unpaid, and partially paid fees.
4. The system updates payment status records.

ALTERNATE FLOWS

- If no invoices exist, the system displays a notification.

POSTCONDITIONS

- Fee payment records are updated and monitored successfully.

12. SEND REMINDERS FOR OVERDUE PAYMENTS

PRIMARY ACTOR

System (Automated)

PRECONDITIONS

- Unpaid invoices must exist.
- Parent or student contact information must be available.

MAIN FLOW

1. The system checks payment status records.
2. The system identifies overdue payments.
3. The system generates reminder notifications.
4. The system sends reminders through notifications or SMS.

ALTERNATE FLOWS

- If contact information is invalid, the system logs the failed notification.

POSTCONDITIONS

- Payment reminders are sent successfully.

13. PERFORM DATA BACKUPS

PRIMARY ACTOR

System (Automated)

PRECONDITIONS

- The database and system storage must be operational.

MAIN FLOW

1. The system initiates the backup process automatically.
2. The system copies data securely.
3. The system stores the backup in a backup location.
4. The system verifies backup completion.

ALTERNATE FLOWS

- If backup storage is unavailable, the system logs an error and retries later.

POSTCONDITIONS

- System data is backed up successfully.

14. SWITCH INTERFACE LANGUAGE (ARABIC / ENGLISH)

PRIMARY ACTOR

Student / Parent / Administrator

PRECONDITIONS

- The user must be logged into the system.

MAIN FLOW

1. The user opens system settings.
2. The user selects the preferred language.
3. The system applies the selected language.
4. The interface reloads with the new language.

ALTERNATE FLOWS

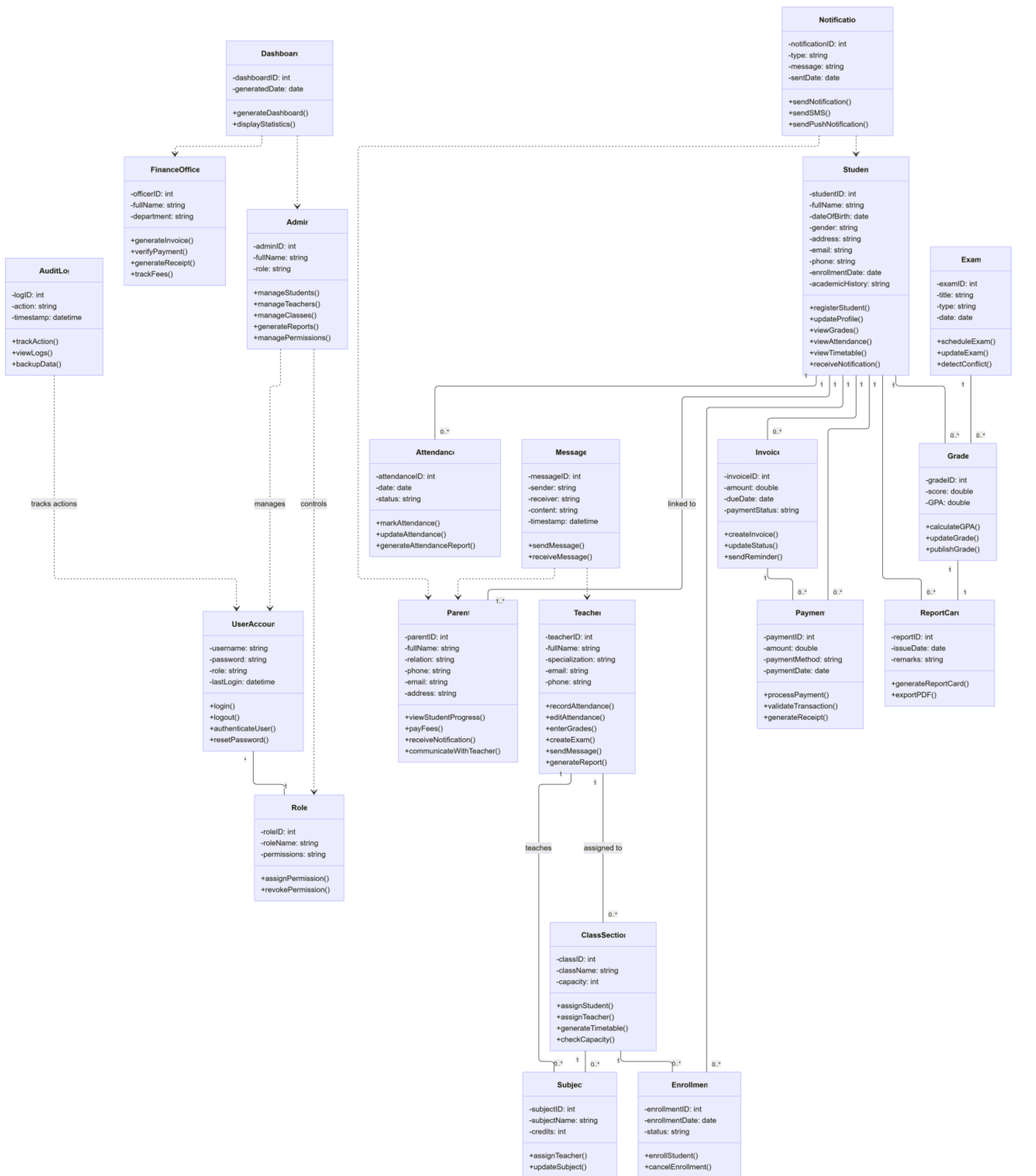
- If the selected language is unavailable, the system keeps the current language.

POSTCONDITIONS

- The interface language is updated successfully.

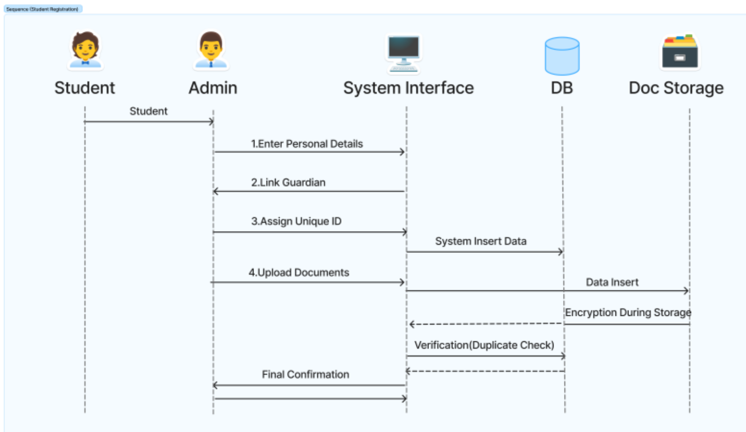
TASK 6

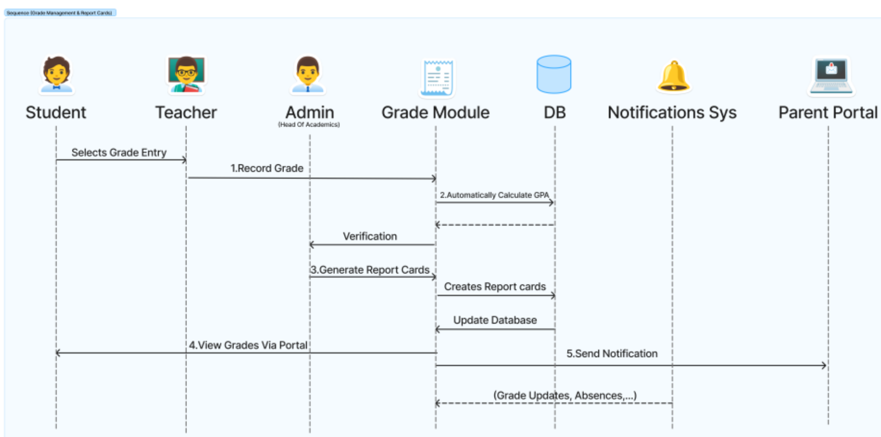
Class Diagram

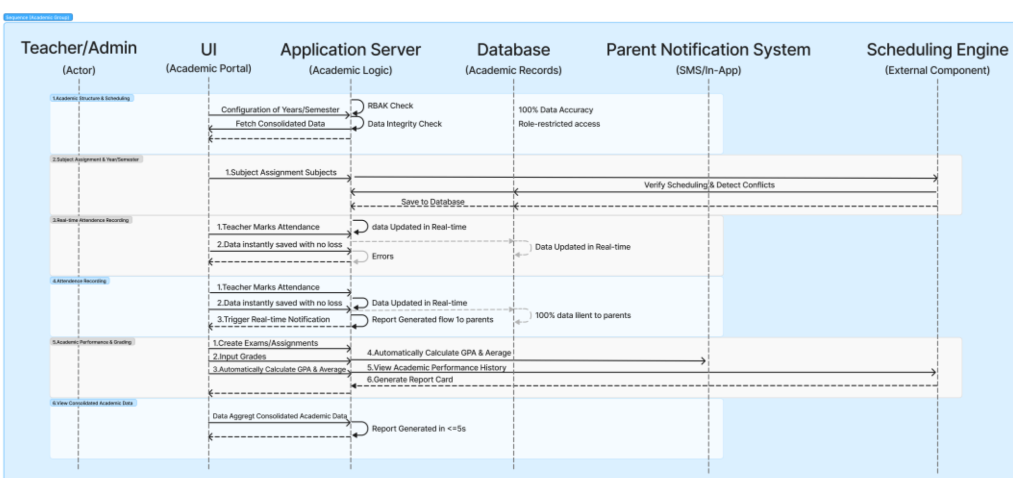


TASK 7

Behavioral Diagram







TASK 8

Assumptions, Constraints & Risks

1. Business Assumptions

- **Data Availability**: It is assumed that all current manual paperwork and Excel sheets at FVIS are accurate and available for digitization.
- **Stakeholder Cooperation**: We assume that teachers, parents, and administrative staff will be willing to participate in the requirements gathering and training phases.
- **Management Support**: There is an assumption that the school board will provide the necessary budget and infrastructure to support the transition to the new Integrated School Management System (ISMS).

2. Technical Constraints

- **Platform Accessibility**: The system must be accessible via standard web browsers and should have a mobile-responsive interface for parents and teachers.
- **Security Standards**: The software must comply with data privacy and access control regulations to protect sensitive student and financial records.
- **Performance**: The system must be able to handle simultaneous access by a growing number of students and staff without significant lag.

3. Project Risks

- **Resistance to Change**: Staff members accustomed to manual processes may resist adopting the new digital system.
- **Data Migration Errors**: There is a risk of losing or corrupting data when moving information from fragmented Excel sheets to the centralized database.
- **Timeline Delays**: The complexity of integrating finance and academic modules may lead to delays in the project delivery schedule.

TASK 9
System Scope Definition

The System Scope Definition establishes the boundaries of the ISMS Version 1.0. It explicitly identifies what the system will and will not include, and provides justification for each scope decision. This document serves as the authoritative scope baseline — no feature may be added or removed without a formal change request reviewed by the full team.

9.1 In-Scope Features

The following features and modules are confirmed for delivery within ISMS Version 1.0. Each item is linked to the business objectives defined in Task 1.

ID	Feature / Module	Description	Linked Objective
S-01	**Student Information Management**	Full digital student registration, profile management, document upload, enrollment history, and search/filter capabilities.	OBJ-01
S-02	**Teacher & Staff Management**	Staff profile creation and management, subject and class assignment, staff directory, and attendance tracking.	OBJ-02
S-03	**Academic Year & Class Configuration**	Creation and management of academic years, semesters, class sections, subjects, and curricula.	OBJ-04
S-04	**Attendance Recording & Notifications**	Digital daily attendance marking by teachers with automatic real-time SMS/push notifications to parents upon student absence.	OBJ-02
S-05	**Grade & Exam Management**	Grade entry for assignments, quizzes, and exams; automatic GPA calculation; exam scheduling with conflict detection; and report card generation.	OBJ-04
S-06	**Fee Structure & Invoice Management**	Configurable fee structures, automated per-student invoice generation per term, partial payment tracking, and overdue flagging.	OBJ-03
S-07	**Online Fee Payment**	Secure online payment via credit/debit card and bank transfer, integrated with a third-party payment gateway supporting EGP transactions.	OBJ-03
S-08	**Parent Portal**	Dedicated web-accessible portal for parents to view child's grades, attendance, schedule, fee status, and communicate with teachers.	OBJ-05

S-09	**Student Portal**	Self-service portal for students to access their grades, timetable, attendance record, and assignment submissions.	OBJ-05
S-10	**Executive Dashboard & Reporting**	Real-time KPI dashboards for the Principal and Board; standard and custom reports exportable in PDF and Excel.	OBJ-06
S-11	**Role-Based Access Control (RBAC)**	Distinct permission sets per role (Admin, Teacher, Student, Parent, Finance, Principal), enforced at the system level.	OBJ-07
S-12	**Audit Logs & Automated Backups**	Immutable audit trail for all data changes; automated daily database backups to secure offsite storage.	OBJ-07
S-13	**Bilingual Interface (Arabic / English)**	Full Arabic (RTL) and English (LTR) language support with a user-toggleable language switch.	OBJ-05

9.2 Out-of-Scope Features

The following features are explicitly excluded from ISMS Version 1.0. These items were considered during scoping discussions but deferred to future releases based on the justifications provided. Including any of these features in Version 1.0 would increase delivery risk, timeline, and cost without addressing the core business problems identified in Task 1.

ID	Excluded Feature	Reason for Exclusion
X-01	**Native Mobile Applications (iOS / Android)**	Version 1.0 will deliver a fully mobile-responsive web interface accessible from any smartphone browser. Native app development requires a separate development pipeline, app store approvals, and significant additional budget. Deferred to Version 2.0.
X-02	**Learning Management System (LMS)**	Features such as live video classes, recorded lecture libraries, interactive course content, and online quizzes go beyond the administrative scope of this project. A dedicated LMS integration may be evaluated as a separate initiative.
X-03	**Ministry of Education Portal Integration**	Integration with external government educational databases is subject to regulatory approval, undefined API availability, and security constraints outside the school's control. Deferred pending official guidelines.
X-04	**AI-Powered Predictive Analytics**	Machine learning models for predicting student performance, dropout risk, or enrollment trends require large historical datasets not yet available in a structured form. Deferred to Version 3.0 after sufficient data is accumulated.
X-05	**Library Management Module**	Book cataloguing, borrowing, and return tracking is a separate operational domain not linked to the core business problems identified. Can be added as a standalone module in a future release.

X-06	**Transportation & GPS Tracking**	School bus routing, GPS tracking, and student pick-up/drop-off management is outside the academic and administrative scope of this project. Requires hardware integration beyond the software boundary.
X-07	**Cafeteria & Inventory Management**	Canteen ordering, inventory tracking, and food service management are operationally independent from student academic management. Not part of the identified business problems.
X-08	**HR Payroll Processing**	Staff salary calculation, payslip generation, and payroll compliance reporting are governed by HR and financial regulations requiring a dedicated payroll system. Outside the scope of a school management system.
X-09	**Third-Party Social Media Integration**	Connecting the ISMS to external platforms (e.g., WhatsApp, Facebook) for notifications or communication is a security risk and is not required given the system's built-in notification and messaging modules.

9.3 Scope Decision Justification & Principles

Guiding Principles

The following principles governed all scope inclusion and exclusion decisions:

- Problem-Driven Inclusion: A feature was included in scope only if it directly addresses one or more of the 8 business problems identified in Task 1. Features without a clear problem link were excluded.
- Feasibility Over Ambition: Features whose implementation would require third-party dependencies outside the school's control (government APIs, hardware, regulatory approvals) were excluded from Version 1.0.
- Core Before Extended: The scope prioritizes getting core operations right — student records, attendance, grades, fees, communication — before adding extended or advanced capabilities.
- Phased Delivery Model: Excluded features are not permanently rejected; they are deferred to future versions (2.0, 3.0) following a structured evaluation after Version 1.0 go-live.
- Stakeholder Consensus: Scope boundaries were determined based on stakeholder interviews and the business problem statement provided by the school board.

Task 9 Summary

Task 9 has established the definitive scope boundary for ISMS Version 1.0. Thirteen features are confirmed in-scope, nine are explicitly out-of-scope with clear justifications, and a three-version phased roadmap has been defined. This scope baseline is locked. Any proposed additions must go through a formal change request reviewed and approved by the full team and project supervisor.